

Evolving Horizons: Exploring System Analysis and Design from Past to Future

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Abstract—This study explores the historical turning points, contemporary patterns, and anticipated future developments in system analysis and design. It starts with a summary of past advancements, moves through modern approaches and resources, and foresees the new paradigms that will influence the profession going forward. The purpose of this investigation is to offer insights into how system analysis and design are developing.

Keywords— System Analysis, System Design, Trends, Evolution, Artificial Intelligence

I. INTRODUCTION

System analysis and design are at the roots of technological development, allowing for the creation of efficient and solid systems across a wide range of areas. This paper takes a journey through time, following the growth of system analysis and design from its inception to its current condition and beyond. We hope to shed light on the ever-changing frontiers of this critical field by reviewing previous accomplishments, current techniques, and prospects.

II. PAST

System analysis and design have their origins in earlier times. The subject has been improved over the years by the emergence of techniques like RAD and SSADM. The basis for current methods was established by these historical changes.

A. SSADM

Structured Systems Analysis and Design Method (SSADM) is a popular computer application development method in the United Kingdom stated in [6]. It is usually stated as an essential for government computing projects. There are many benefits of using SSADM to build a system. The

benefits are improving project management and control, make effective use for experienced and unexperienced employee, develop better system, and provide a structure for effective communication amongst project participants. Therefore, SSADM an essential tool because it improves project management, gives experienced and inexperienced staff more active, and encourages the development of better systems and effective communication between all project stakeholders.

B. RAD

RAD or Rapid Application Development is an adaptable method for quickly deploying software applications. This approach is designed to efficiently accommodate changes and incorporate new inputs such as features and updates throughout the development process. RAD consists of five main phases which are application development, testing and turnover, data and business modelling, and process modelling. With RAD approach, development cycles are faster, products are built based on client needed and incremental progress is more visible and measurable.

III. PRESENT

System analysis and design are currently experiencing a paradigm change because of technical breakthroughs and shifting market demands. Agile approaches have gained popularity, focusing on iterative development and customer participation. The function of agile approaches as the guide is for developer team react swiftly to client feedback or changes in the market without throwing off a year's worth of plans. Other than that,

DevOps approaches have transformed the integration of development and operations, simplifying the software delivery process. Furthermore, the rise of cloud computing and microservices architectures has fundamentally altered system design concepts, allowing for unprecedented scalability and adaptability.

IV. FUTURE

If we look at the future, the changing of technology leads to the change of system analysis and design. As today we have Artificial Intelligence that might affect system analysis and design soon in the future. AI are transforming the way system analyst and design collect, process and interpret data. Large data sets can be analyzed by AI to find patterns, trends, and insights that conventional approaches can miss. Not to mention, AI can assist in requirements gathering by analyzing large datasets, user feedback, and historical project data to identify patterns, preferences, and potential conflicts. Natural Language Processing (NLP) techniques can help extract requirements from unstructured text, such as user stories, emails, or documentation. In this case, Machine Learning (ML), plays a crucial role in NLP, enabling systems to understand, interpret, and generate human language. Moreover, AI may assist in producing architectural diagrams, design mock-ups, and prototypes based on user requirements and design concepts. Algorithms for generative design can explore a vast array of design options and optimizing solutions in accordance with specified parameters and restrictions. With the power tools that AI have, it can automatically and starting to code, debug, and test. It enhances the testing process. It includes the automation test of generating, executing and the resulting analysis. If we talk about the deployment, AI can ensure consistency, reliability, and scalability across different deployment environments by analyzing historical performance data, user feedback, and environmental factors. Lastly, AI can also perform maintenance. It can detect anomalies and security threats in real-time with the monitoring system

which can lead to proactive maintenance and issue resolution.

V. CONCLUSION

The evolution of system analysis and design has been marked by significant advancements from its inception to the present day, with exciting prospects for the future. The framework for modern methods was laid in the past with the introduction of foundational methodologies like RAD and SSADM. Meanwhile in the present, Agile methodologies and DevOps approaches have become mainstream, emphasizing iterative development, customer collaboration, and seamless integration of development and operations. Not to be left that AI will undoubtedly influence system analysis and design. AI can propel system analysis and design into a new era of efficiency, agility, and creativity.

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